



COMPETENCY STANDARD

FOR

ADVANCED IMAGE PROCESSING & TECHNIQUE

(ICT SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Advance Image Processing & Technique serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the ICT sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for Advance Image Processing & Technique. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (20 hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-ICT-AIPT-01-G	Demonstrate Interpersonal and Communication Skills	1 Apply interpersonal skills at the workplace 2 Understand and communicate information and ideas 3 Prepare and present reports and workplace documents	10
SICIP-ICT-AIPT-02-G	Support Innovation and Manage Change	1 Identify needs for innovation in the workplace 2 Apply creative approach and solution 3 Support flexible and innovative ways of working 4 Adapt to emerging technological changes and opportunities	10
Total Hour			20 Hrs.

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-ICT-AIPT-01-S	Comply with Ethical Standards in IT Workplace	1. Uphold the requirements of clients 2. Deliver quality products and services 3. Maintain professionalism at workplace 4. Maintain workplace code of conduct	20
Total Hour			20 Hrs.

Occupation Specific Competencies (180 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-ICT-AIPT-01-O	Apply Basic Graphic Design Fundamentals	1. Identify basic design elements and principles 2. Apply color theory and typography basics 3. Use layout and alignment techniques 4. Create simple visual compositions 5. Operate basic graphic design software tools	15
SICIP-ICT-AIPT-02-O	Apply Visual Design and Image Editing Fundamentals	1. Apply basic visual design principles in image work 2. Use color balance and tone adjustments 3. Perform basic image editing tasks using image editing tools 4. Save and export images in appropriate formats	25
SICIP-ICT-AIPT-03-O	Perform Image Composition and Editing Automation	1. Apply image composition techniques 2. Perform advanced image editing operations 3. Use layers, masking, and blending techniques 4. Apply automation and batch processing tools 5. Organize and manage multiple image files efficiently	45
SICIP-ICT-AIPT-04-O	Perform Advanced Image Editing and Retouching	1. Retouch images for professional output 2. Perform detailed color correction and grading 3. Remove background and unwanted objects 4. Optimize images for web and print use 5. Save and transfer images according to client requirements	45
SICIP-ICT-AIPT-05-O	Apply AI-Based Image Generation and Processing	1. Identify AI-based image generation and processing. 2. Generate images using AI-based techniques 3. Process and enhance images using AI tools 4. Apply quality and ethical standards in AI image generation 5. Organize and prepare AI-generated images for use	25
SICIP-ICT-AIPT-06-O	Perform Image Quality Assurance and Documentation	1. Apply image quality standards and guidelines 2. Perform quality checking and error correction	10

Code	Unit of Competency	Elements of Competency	Duration (hours)
		3. Follow quality assurance procedures 4. Maintain documentation and version control 5. Report quality issues and corrective actions	
SICIP-ICT-AIPT-07-O	Develop Freelancing and Outsourcing Career	1. Identify freelancing and outsourcing market opportunities 2. Develop professional profile and service portfolio 3. Prepare proposals, gigs, and acquire clients 4. Manage client communication and long-term relationships 5. Plan career growth and continuous professional development	15
Total Hour			0 Hrs.

COMPETENCY STANDARD: ADVANCE IMAGE PROCESSING & TECHNIQUE

The Generic Competencies

Unit of Competency: DEMONSTRATE INTERPERSONAL AND COMMUNICATION SKILLS	Nominal Duration: 10 hrs.	Unit Code: SICIP-ICT-AIPT-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitude required to demonstrate interpersonal and communication skills. It specifically includes the tasks of applying interpersonal skills at the workplace, understanding and communicating information and ideas, preparing and presenting reports and workplace documents.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply interpersonal skills at the workplace	1.1 <u>Workplace interpersonal skills</u> are interpreted. 1.2 The impacts of interpersonal skills are described. 1.3 Interpersonal skills are performed in accordance with workplace procedures. 1.4 <u>Ways of improving interpersonal skills</u> are identified and followed.
2. Understand and communicate information and ideas	2.1 <u>Communication tools and equipment</u> are operated, and faults are identified and reported. 2.2 Consultative processes is used to collect and convey information. 2.3 Information systems is used to store, retrieve and update information. 2.4 Information data and information are analyzed and interpreted. 2.5 Information is communicated with individuals and groups.
3. Prepare and present reports and workplace documents	3.1 <u>Workplace documents</u> are interpreted. 3.2 Appropriate format for document is used to meet workplace requirements. 3.3 Document is drafted in accordance with workplace procedures. 3.4 Document is edited and presented in a final version appropriate to task.

Range of variables:

Variables	Range (may include but not limited to)
1. Interpersonal skills	1.1 Effective communication 1.2 Cooperation

	1.3 Giving and receiving feedback, feed-forward and professional criticism 1.4 Pressure handling 1.5 Time management 1.6 Acceptance of diversity 1.7 Supporting and encouraging others 1.8 Inclusiveness 1.9 Problem solving 1.10 Negotiation 1.11 Conflict management
2. Ways of improving interpersonal skills	1.1 Cultivate a positive outlook 1.2 Ask trusted friends or colleagues for constructive criticism 1.3 Control emotions 1.4 Observe other positive interpersonal interactions 1.5 Practice active listening 1.6 Seek out opportunities to build relationships 1.7 Practice empathy
3. Communication tools and equipment	1.1 Telephone 1.2 Internet 1.3 Mobile Phone 1.4 Fax machines 1.5 Two-way radio
4. Workplace documents	1.1 Business Letters 1.2 Business Reports 1.3 Transactional Documents 1.4 Memos 1.5 Forms 1.5.1 Memorandum 1.5.2 Requisitioning Form 1.5.3 Personnel Form 1.6 Safety Report Form

Curricular Content Guide

1. Underpinning knowledge	1.1 Interpersonal skills 1.2 Impacts of interpersonal skills 1.3 Consultative processes 1.4 Workplace documents
2. Underpinning skill	1.5 Demonstrating interpersonal skills 1.6 Collecting workplace information 1.7 Analyzing information and ideas 1.8 Preparing workplace documents 1.9 Presenting workplace documents
3. Underpinning Attitudes	1.10 Commitment to occupational health and safety practices 1.11 Promptness in carrying out activities 1.12 Tidiness and timeliness 1.13 Respect to peers, sub-ordinates and seniors in workplace

	1.14 Environmental concern 1.15 Sincerity and honesty
4. Resource implications	The following resources must be provided: 1.1 Workplace (actual or simulated). 1.2 Relevant materials and equipment. 1.3 Relevant specifications or work instructions.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 applied interpersonal skills at the workplace 1.2 communicated information and ideas 1.3 prepared and presented reports and workplace documents
2. Methods of assessment	Competency should be assessed by: 1.4 Written test 1.5 Practical Demonstration 1.6 Oral Questioning 1.7 Portfolio (Optional)
3. Context of assessment	1.8 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.9 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: SUPPORT INNOVATION AND MANAGE CHANGE	Nominal Duration: 10 hrs.	Unit Code: SICIP-ICT-AIPT-02-G
Unit Descriptor: This unit covers the knowledge, skills, and attitudes required to support innovation and manage change. It specifically includes the tasks of identifying needs for innovation in the workplace, applying creative approaches and solution, supporting flexible and innovative ways of working and adapting to emerging technological changes and opportunities.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify needs for innovation in the workplace	1.1. The need for <u>innovation</u> in scope of work is recognized 1.2. The value of <u>innovative practices</u> in the workplace is recognized 1.3. Existing way of working in the workplace are interpreted. 1.4. Drawbacks of existing way of working are identified. 1.5. Benefits of change are identified to make it consistent with organizational requirements 1.6. Realistic timelines and targets for implementation of changes are set.
2. Apply creative approach and solution	2.1. Opportunities and creative approaches to implement innovation practices are identified 2.2. Creative approaches of coworkers pertaining to work practices are analyzed and incorporated 2.3. Innovation in accordance with organizational requirements changes are implemented
3. Support flexible and innovative ways of working	3.1. Maximum innovative opportunities are promoted. 3.2. Work assignments to facilitate innovative work skills are organized. 3.3. Team members are provided with guidance and coaching on innovation in the workplace. 3.4. Models of innovative work practice are provided and discussed 3.5. Appropriate environment for learning and innovation is maintained.
4. Adapt to emerging technological changes and opportunities	4.1. Usages of different <u>technologies</u> is determined based on job requirements. 4.2. Appropriate technology is selected as per work specifications. 4.3. Relevant technology is effectively used in carrying out functions.

	4.4. Appropriate implementation tools are used as per task requirement.
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Range of variables:

Variables	Range (may include but not limited to)
1. Innovative practices	1.1. Self-directed support to implement new ideas 1.2. Collaborative arrangement to apply new method of working 1.3. Making scope of work more efficient to use new technology
2. Innovation	1.4. New ideas 1.5. Different ideas 1.6. New methods of doing work 1.7. Use of new Technology
3. Technologies	1.8. Office technology 1.9. Industrial technology 1.10. System technology 1.11. Information technology 1.12. Training technology

Curricular Content Guide

1. Underpinning knowledge	1.1. Definition of innovation 1.2. Current practice in own scope of work 1.3. Workplace procedures 1.4. Support required to generate creative ideas 1.5. Difference between innovation and creativity 1.6. Common effects of change and innovation in the workplace 1.7. Industrial and organizational context of change 1.8. Organization's policies, plans, procedures and structure 1.9. Knowledge of resources required by the organization's operations 1.10. Awareness on technology and its function 1.11. Operating instructions 1.12. Applicable software 1.13. Company policy in relation to relevant technology 1.14. Technology adaptability
2. Underpinning skill	1.1. Identifying resources required for creativity and innovation 1.2. Contributing to brainstorming session 1.3. Identifying issues and concerns of one's scope of work 1.4. Encouraging co-workers to generate and develop ideas 1.5. Evaluating potential obstacles to and opportunities for creativity and innovation 1.6. Sharing of best practices related to innovation and creativity 1.7. Applying relevant technology 1.8. Using update machine and Software

	1.9. Acquiring troubleshooting skills Identifying issues 1.10. Identifying current industry standard diagnostic tools 1.11. Describing common malfunctions and resolutions. 1.12. Determining the root cause of a routine malfunction
3. Underpinning Attitudes	1.13. Commitment to occupational health and safety 1.14. Promptness in carrying out activities 1.15. Sincere and honest to duties 1.16. Eagerness to learn 1.17. Tidiness and timeliness 1.18. Environmental concerns 1.19. Respect for rights of peers and seniors at workplace 1.20. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 1.21. Workplace (actual or simulated). 1.22. Relevant materials and equipment. 1.23. Relevant specifications or work instructions.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified need for innovation in the area of work 1.2 applied creative approach and solution 1.3 supported flexible and innovative ways of working 1.4 adapted to emerging technological changes and opportunities
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: COMPLY WITH ETHICAL STANDARDS IN IT WORKPLACE	Nominal Duration: 20 hrs.	Unit Code: SICIP-ICT-AIPT-01-S
Unit Descriptor: This unit covers knowledge, skills and attitudes required to comply with ethical standards in IT workplace. It specifically includes the tasks of upholding the requirements of clients, delivering quality products and services, maintaining professionalism at workplace and maintaining workplace code of conduct.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Uphold the requirements of clients	1.1 Clients' requirements are identified. 1.2 Confidentiality of information is maintained in accordance with workplace policies/organizational policies/national legislation. 1.3 Potential conflicts of interest are identified and involved parties of potential conflicts are notified. 1.4 Proprietary rights of client/customer are asserted.
2. Deliver quality products and services	2.1. Products and services are provided according to the clients' requirements. 2.2. Work is completed as per standards. 2.3. Quality processes are implemented when developing products and services.
3. Maintain professionalism at workplace	3.1 Work processes are delivered as per standards. 3.2 Skills, knowledge and qualifications are presented in a professional manner. 3.3 Services and products developed by self and others are delivered as per workplace standard. 3.4 Unbiased and objective information are provided to the clients. 3.5 Realistic estimates for time, cost and delivery of outputs are presented during negotiation.
4. Maintain workplace code of conduct	4.1 Workplace code of conduct is interpreted. 4.2 Workplace code of conduct is followed.

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Corporate code of confidentiality of information 1.2 Organizational policies, national legislation and workplace policies in relation to IT sector 1.3 Law and regulations pertaining to proprietary rights 1.4 Quality processes for products and services
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	1.5 Procedure of provided to client information 1.6 Method of estimating for time, cost and delivery products and services 1.7 Workplace code of conduct in IT sector
2. Underpinning Skills	2.1 Upholding confidentiality of information in accordance with organizational policies, national legislation and workplace policies 2.2 Asserting proprietary rights of clients/customers 2.3 Completing work in accordance with industry and international standards 2.4 Implementing quality processes when developing products and services 2.5 Delivering correctly services and products developed by self and others 2.6 Providing unbiased and objective information to the clients 2.7 Presenting realistic estimates for time, cost and delivery of outputs during negotiation 2.8 Maintaining workplace code of conduct
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 National legislation in relation to IT sector 4.4 Workplace policies, code of conduct & proprietary rights of clients in IT sector 4.5 Relevant tools, equipment and facilities needed to perform the activities 4.6 Required learning materials

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 upheld the requirements of clients/customers 1.2 asserted proprietary rights of clients/customers 1.3 delivered quality products and services 1.4 Implemented quality processes when developing products and services 1.5 maintained professionalism at the workplace 1.6 completed work according to industry and international standards
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	1.7 maintained workplace code of conduct
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: APPLY BASIC GRAPHIC DESIGN FUNDAMENTALS	Nominal Duration: 15 hrs.	Unit Code: SICIP-ICT-AIPT-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply basic graphic design fundamentals. It specifically includes the tasks of identifying basic design elements and principles, applying color theory and typography basics, using layout and alignment techniques, creating simple visual compositions and operating basic graphic design software tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify basic design elements and principles	1.1 <u>Basic design elements</u> are identified. 1.2 <u>Design principles</u> and proportion are recognized. 1.3 Relationship between design elements and principles is explained. 1.4 Appropriate elements and principles are selected for simple design tasks.
2. Apply color theory and typography basics	2.1 <u>Basic color theory concepts</u> are identified and applied in design work. 2.2 Color combinations are selected based on harmony and contrast. 2.3 <u>Appropriate font types</u> are selected according to design purpose. 2.4 <u>Typography elements</u> are applied correctly. 2.5 Text and color are combined to ensure readability and visual appeal.
3. Use layout and alignment techniques	3.1 <u>Appropriate layout structures</u> are selected for the design task. 3.2 Design elements are arranged using proper alignment techniques. 3.3 Spacing and margins are applied to ensure visual balance. 3.4 Grids and guides are used to maintain consistency. 3.5 Overall layout is reviewed for clarity and organization.
4. Create simple visual compositions	4.1 Basic design elements are combined to form simple visual compositions. 4.2 Appropriate color and typography are applied in the composition. 4.3 <u>Layout and alignment principles</u> are maintained throughout the design.

	4.4 Visual balance and proportion are ensured in the composition. 4.5 Final composition is reviewed for clarity and visual appeal.
5. Operate basic image processing software tools	5.1 <u>Basic image processing software tools</u> are identified. 5.2 Image processing software interfaces are accessed and navigated. 5.3 <u>Selection tools</u> are identified and operated. 5.4 <u>Pen tools</u> are identified and operated for drawing paths. 5.5 Crop and transformation tools are identified and operated. 5.6 Image adjustment tools are recognized and operated.

Range of Variables

Variable	Range May include but not limited to:
1. Basic design elements	1.1 Line 1.2 Shape 1.3 Color 1.4 Texture 1.5 Space
2. Design principles	2.1 Balance 2.2 Contrast 2.3 Alignment 2.4 Proportion 2.5 Rhythm 2.6 Hierarchy
3. Basic color theory concepts	3.1 Primary Colors 3.2 Secondary Colors 3.3 Tertiary Colors 3.4 Color Wheel 3.5 Warm Colors 3.6 Cool Colors 3.7 Hue 3.8 Tint 3.9 Shade 3.10 Tone
4. Appropriate font types	4.1 Serif Fonts 4.2 Sans Serif Fonts 4.3 Script Fonts 4.4 Display Fonts 4.5 Monospace Fonts
5. Typography elements	5.1 Size 5.2 Spacing 5.3 Alignment 5.4 Tracking

	5.5 Leading
6. Appropriate layout structures	6.1 Single Column Layout 6.2 Multi-Column Layout 6.3 Grid Layout 6.4 Modular Layout 6.5 Hierarchical Layout 6.6 Z-Pattern Layout 6.7 F-Pattern Layout 6.8 Rules of third
7. Layout and alignment principles	7.1 Balance 7.2 Alignment 7.3 Proximity 7.4 Repetition 7.5 Contrast 7.6 White Space 7.7 Visual Hierarchy 7.8 Consistency
8. Basic image processing software tools	8.1 Adobe Photoshop 8.2 Adobe Lightroom 8.3 Canva 8.4 Microsoft Paint
9. Selection tools	9.1 Marquee tools 9.2 Lasso tools 9.3 Quick selection tools 9.4 Magic wand tools 9.5 Object selection tools
10. Pen tools	10.1 Pen tool 10.2 Freeform pen tool 10.3 Curvature pen tool 10.4 Add anchor point tool 10.5 Delete anchor point tool 10.6 Convert point tool

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concept of basic design elements and principles and their role in visual design. 1.2 Identify design elements and principles such as balance and proportion. 1.3 Understand relationship between design elements and principles in creating effective designs. 1.4 Concept of color theory and typography basics for visual communication. 1.5 Identify color harmony, contrast, and font types for
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	readability and appeal. 1.6 Concept of creating visual compositions using elements, color, and typography. 1.7 Understand visual balance, proportion, and clarity in design outputs. 1.8 Identify basic image processing tools and software functions for editing and design tasks.
2. Underpinning Skills	2.1 Identifying and selecting basic design elements and principles for design tasks. 2.2 Applying color theory and selecting appropriate color combinations. 2.3 Selecting and using suitable font types and typography elements for readability. 2.4 Applying layout, alignment, spacing, and grid techniques in design work. 2.5 Arranging and organizing design elements to create balanced compositions. 2.6 Using basic image processing software interfaces and tools effectively. 2.7 Operating selection, pen, crop, and transformation tools for design tasks. 2.8 Adjusting and refining images using basic image editing and adjustment tools.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Relevant tools 4.4 Facilities needed to perform the activities 4.5 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 identified basic design elements and principles 1.2 applied color theory and typography basics 1.3 used layout and alignment techniques
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	1.4 created simple visual compositions 1.5 operated basic graphic design software tools
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY VISUAL DESIGN AND IMAGE EDITING FUNDAMENTALS	Nominal Duration: 25 hrs.	Unit Code: SICIP-ICT-AIPT-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply visual design and image editing fundamentals. It specifically includes the tasks of applying basic visual design principles in image work, using color balance and tone adjustments, performing basic image editing tasks using image editing tools and saving and exporting images in appropriate formats.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply basic visual design principles in image work	1.1 Basic visual design principles are applied in image editing tasks. 1.2 Balance, contrast, and alignment are maintained in image compositions. 1.3 Color harmony and proportion are ensured in visual work. 1.4 Focal points are created to enhance visual impact.
2. Use color balance and tone adjustments	2.1 <u>Color balance settings</u> are adjusted to achieve accurate color representation. 2.2 Brightness and contrast levels are modified as required. 2.3 Tone adjustments are applied to improve image clarity. 2.4 Highlights and shadows are corrected to enhance visual quality. 2.5 Final color and tone outputs are reviewed for consistency and accuracy.
3. Perform basic image editing tasks using image editing tools	3.1 <u>Appropriate image editing tools</u> are selected according to task requirements. 3.2 Images are cropped, resized, rotated, and adjusted as required. 3.3 Brightness, contrast, and basic corrections are applied using suitable tools. 3.4 Unwanted elements are removed and simple retouching is performed. 3.5 <u>Tool settings</u> are adjusted to achieve desired results. 3.6 Edited images are reviewed to ensure quality and accuracy.
4. Save and export images in appropriate formats	4.1 Appropriate file formats are selected based on usage requirements. 4.2 Image resolution and quality settings are adjusted

	before saving. 4.3 Files are saved using proper naming conventions. 4.4 Images are exported in <u>suitable formats</u> for print or digital use. 4.5 Exported files are verified to ensure quality and compatibility.
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Range of Variables

Variable	Range May include but not limited to:
1. Color balance settings	1.1 Brightness Adjustment 1.2 Contrast Adjustment 1.3 Hue/Saturation Adjustment 1.4 Temperature (Warm/Cool) 1.5 Tint Adjustment 1.6 Exposure Control 1.7 Curve Adjustment
2. Appropriate image editing tools	2.1 Crop Tool 2.2 Resize Tool 2.3 Rotate Tool 2.4 Selection Tool 2.5 Brush Tool 2.6 Eraser Tool 2.7 Clone/Stamp Tool 2.8 Healing Tool 2.9 Text Tool 2.10 Color Adjustment Tool 2.11 Content Aware Tool
3. Tool settings	3.1 Brush tip preview 3.2 Brush angle and roundness 3.3 Brush size 3.4 Brush hardness 3.5 Opacity 3.6 Flow 3.7 Feather 3.8 Tolerance
4. Suitable formats	4.1 .JPEG (JPG) 4.2 .PNG 4.3 .GIF 4.4 .PDF 4.5 .PSD

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concept of visual design principles such as balance, contrast, alignment, and focal point. 1.2 Understand application of color harmony, proportion, and visual impact in image work. 1.3 Concept of color balance, brightness, contrast, and tone adjustments in images. 1.4 Understand highlights, shadows, and tonal correction techniques for image quality. 1.5 Identify image editing tools and their functions for basic editing tasks. 1.6 Understand retouching techniques and removal of unwanted elements. 1.7 Concept of file formats, resolution, and quality settings for different uses. 1.8 Procedures for saving, naming, and exporting images for print and digital media.
2. Underpinning Skills	2.1 Applying basic visual design principles in image editing tasks. 2.2 Maintaining balance, contrast, alignment, and focal points in compositions. 2.3 Adjusting color balance, brightness, contrast, and tone for image quality. 2.4 Correcting highlights and shadows to enhance visual clarity. 2.5 Selecting and using appropriate image editing tools for different tasks. 2.6 Removing and refining unwanted elements using editing tools. 2.7 Selecting suitable file formats, resolution, and quality settings. 2.8 Saving and exporting images with proper naming and verifying output quality.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Multimedia

	4.4 Tools (content writing, graphic design, video creation) 4.5 Required learning materials 4.6 Specifications and work instructions
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 applied basic visual design principles in image work 1.2 used color balance and tone adjustments 1.3 performed basic image editing tasks using image editing tools 1.4 saved and exported images in appropriate formats
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM IMAGE COMPOSITION AND EDITING AUTOMATION	Nominal Duration: 45 hrs.	Unit Code: SICIP-ICT-AIPT-03-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to perform image composition and editing automation. It specifically includes the tasks of applying image composition techniques, performing advanced image editing operations, using layers, masking and blending techniques, applying automation and batch processing tools and organizing and managing multiple image files efficiently.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply image composition techniques	1.1 <u>Image composition principles</u> are identified and applied. 1.2 Multiple images and graphic elements are arranged to create a balanced composition. 1.3 <u>Layers and blending techniques</u> are used to combine visual elements. 1.4 Foreground and background elements are adjusted to achieve visual harmony. 1.5 Final compositions are reviewed to ensure clarity, balance, and visual impact.
2. Perform advanced image editing operations	2.1 <u>Advanced image editing tools</u> and features are identified and used. 2.2 Complex image adjustments and enhancements are applied as per requirement. 2.3 <u>Layer management tools</u> are used to refine image edits as per requirement. 2.4 <u>Advanced retouching techniques</u> are applied as per requirement. 2.5 Edited images are reviewed to ensure quality and consistency.
3. Apply automation and batch processing tools	3.1 <u>Automation features and batch processing tools</u> are identified within the software. 3.2 Actions or automated editing sequences are created and configured. 3.3 Batch processing settings are defined for multiple images. 3.4 Automated processes are applied to perform repetitive editing tasks. 3.5 Processed images are reviewed to ensure consistency and accuracy.
4. Organize and manage	4.1 Multiple image files are organized using appropriate

multiple image files	folder structures.
	4.2 File naming conventions are applied for easy identification.
	4.3 Image files are categorized and sorted according to project requirements.
	4.4 Metadata and file properties are reviewed and updated where required.
	4.5 Managed image files are stored and maintained for easy access and retrieval.

Range of Variables

Variable	Range May include but not limited to:
1. Image composition principles	1.1 Rule of thirds 1.2 Alignment 1.3 Contrast 1.4 Framing 1.5 Leading lines 1.6 Proportion and scale
2. Layers and blending techniques	2.1 Layer visibility and locking 2.2 Layer opacity and fill adjustment 2.3 Layer filter 2.4 Clipping masks 2.5 Layer grouping and arrangement
3. Advanced image editing tools	3.1 Content-aware fill 3.2 Healing and retouching tools 3.3 Clone stamp tools 3.4 Liquify tools 3.5 Smart objects and smart filters 3.6 Perspective and warp tools 3.7 Gradient and color grading tools 3.8 Noise reduction and sharpening tools
4. Layer management tools	4.1 Layers 4.2 Layer styles 4.3 Masks 4.4 Blending modes 4.5 Layer adjustment
5. Advanced retouching techniques	5.1 Skin smoothing and blemish removal 5.2 Frequency separation 5.3 Dodge and burn technique 5.4 Clone and healing retouching 5.5 Eye and teeth enhancement 5.6 Hair refining 5.7 Background cleanup and object removal 5.8 Color and tone correction

	5.9 High-end portrait retouching
6. Automation features and batch processing tools	6.1 Action recording and playback 6.2 Batch image processing 6.3 Automated resizing and renaming 6.4 Script-based automation 6.5 Preset and template application 6.6 Bulk format conversion 6.7 Watermark automation 6.8 Folder-based processing workflows 6.9 Export automation settings 6.10 Metadata and file information automation

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concept of image composition principles such as balance, layering, and visual hierarchy. 1.2 Understand combining multiple images and elements using layers and blending techniques. 1.3 Concept of foreground and background adjustment for visual harmony. 1.4 Identify advanced image editing tools and features for complex editing tasks. 1.5 Concept of automation and batch processing in image editing software. 1.6 Procedures for creating actions and applying automated editing sequences. 1.7 Understand batch processing for handling multiple images efficiently. 1.8 Concept of file organization, naming, metadata, and image management systems.
2. Underpinning Skills	2.1 Applying image composition principles to create balanced visual designs. 2.2 Arranging and combining multiple images and elements using layers and blending techniques. 2.3 Adjusting foreground and background elements to achieve visual harmony. 2.4 Using advanced image editing tools for complex adjustments and enhancements. 2.5 Managing layers and performing advanced retouching techniques. 2.6 Creating and configuring automation actions for repetitive editing tasks. 2.7 Applying batch processing tools to handle multiple images efficiently. 2.8 Reviewing and ensuring consistency and quality of processed images. 2.9 Organizing and managing image files using proper

	naming, structure, and metadata.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Specifications and work instructions 4.4 Learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 applied image composition techniques 1.2 performed advanced image editing operations 1.3 used layers, masking, and blending techniques 1.4 applied automation and batch processing tools 1.5 organized and managed multiple image files efficiently
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM ADVANCED IMAGE EDITING AND RETOUCHING	Nominal Duration: 45 hrs.	Unit Code: SICIP-ICT-AIPT-04-O
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Unit Descriptor:

This unit covers the knowledge, skills and attitudes required to perform advanced image editing and retouching. It specifically includes the tasks of retouching images for professional output, performing detailed color correction and grading, removing background and unwanted objects, optimizing images for web and print use and saving and transferring images according to client requirements.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Retouch images for professional output	1.1 <u>Image retouching tools and techniques</u> are identified and applied. 1.2 Skin, texture, and surface imperfections are corrected using appropriate tools. 1.3 Color, tone, and lighting are adjusted to improve visual quality. 1.4 Fine details are refined to achieve a professional appearance. 1.5 Retouched images are reviewed to ensure high-quality output.
2. Perform detailed color correction and grading	2.1 <u>Color correction tools</u> are identified and applied to adjust image colors. 2.2 White balance, exposure, and color levels are adjusted for accuracy. 2.3 <u>Color grading techniques</u> are applied to achieve the desired visual tone. 2.4 Highlights, shadows, and midtones are adjusted to enhance image quality. 2.5 Corrected and graded images are reviewed to ensure visual consistency.
3. Remove background and unwanted objects	3.1 Background areas are identified and selected using appropriate tools. 3.2 Backgrounds are removed using selection and masking techniques. 3.3 Unwanted objects are identified within the image. 3.4 <u>Object removal tools</u> are applied to eliminate unwanted elements. 3.5 Edited images are refined to ensure natural and seamless results.
4. Optimize images for web and print use	4.1 Image resolution and dimensions are adjusted according to web or print requirements. 4.2 <u>Color modes</u> are converted for appropriate output

	formats. 4.3 File sizes are optimized while maintaining image quality. 4.4 Images are exported in suitable formats for web and print use. 4.5 Optimized images are reviewed to ensure clarity and output quality.
5. Save and transfer images according to client requirements	5.1 Client requirements for file format, resolution, and size are identified. 5.2 Images are saved using appropriate formats and settings. 5.3 File naming conventions are applied according to project requirements. 5.4 Images are transferred to clients through appropriate digital platforms. 5.5 Delivered files are verified to ensure quality and compliance with client specifications.

Range of Variables

Variable	Range May include but not limited to:
1. Image retouching tools and techniques	1.1 Healing brush and spot healing tools 1.2 Clone stamp tool 1.3 Patch tool 1.4 Red-eye correction 1.5 Blemish and wrinkle removal 1.6 Dodge and burn technique 1.7 Noise reduction and sharpening
2. Color correction tools	2.1 Brightness and contrast adjustment 2.2 Levels adjustment 2.3 Curves adjustment 2.4 Hue and saturation adjustment 2.5 Color balance adjustment 2.6 White balance correction 2.7 Selective color adjustment 2.8 Vibrance adjustment 2.9 Color lookup and color grading tools
3. Color grading techniques	3.1 Tone curve adjustment 3.2 Split toning 3.3 Color balance grading 3.4 Cinematic color grading 3.5 Warm and cool tone adjustment 3.6 LUT application 3.7 Selective color grading 3.8 Shadow, midtone, and highlight grading 3.9 Gradient color grading 3.10 Creative color styling
4. Object removal tools	4.1 Clone stamp tool

	4.2 Healing brush tool 4.3 Spot healing brush tool 4.4 Content-aware fill 4.5 Eraser tool 4.6 Magic eraser tool 4.7 Background eraser tool 4.8 Content-aware move tool
5. Color modes	5.1 RGB color mode 5.2 CMYK color mode 5.3 Grayscale mode 5.4 Bitmap mode 5.5 Indexed color mode 5.6 Lab color mode 5.7 Duotone mode

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concept of professional image retouching techniques for improving visual quality. 1.2 Understand correction of skin, texture, and surface imperfections using editing tools. 1.3 Concept of color correction, white balance, exposure, and color grading. 1.4 Understand adjustment of highlights, shadows, and midtones for image enhancement. 1.5 Concept of seamless editing and natural-looking image refinement. 1.6 Understand image optimization for web and print, including resolution and color modes. 1.7 Procedures for file size optimization and exporting images in suitable formats. 1.8 Understand saving, naming, and transferring images according to client requirements.
2 Underpinning Skills	2.1 Applying image retouching techniques to achieve professional-quality outputs. 2.2 Correcting skin, texture, and surface imperfections using appropriate tools. 2.3 Adjusting color, tone, lighting, and exposure for enhanced visual quality. 2.4 Performing detailed color correction and color grading techniques. 2.5 Removing backgrounds and unwanted objects using selection and masking tools. 2.6 Optimizing images for web and print by adjusting resolution, size, and color modes. 2.7 Saving and exporting images in appropriate formats

	based on client requirements. 2.8 Transferring and verifying final image files to ensure quality and compliance.
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4 Resource Implications	The Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Job specifications and work instructions 4.4 Relevant tools, equipment and facilities 4.5 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 retouched images for professional output 1.2 performed detailed color correction and grading 1.3 removed background and unwanted objects 1.4 optimized images for web and print use 1.5 saved and transferred images according to client requirements
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY AI-BASED IMAGE GENERATION AND PROCESSING	Nominal Duration: 25 hrs.	Unit Code: SICIP-ICT-AIPT-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to apply AI-based image generation and processing. It specifically includes the tasks of identifying AI-based image generation and processing, generating images using AI-based techniques, processing and enhancing images using AI tools, applying quality and ethical standards in AI image generation and organizing and preparing AI-generated images for use.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify AI-based image generation and processing tools	1.1 <u>Different AI image generation tools and platforms</u> are identified. 1.2 Features and capabilities of AI-based image tools are reviewed. 1.3 Appropriate tools are selected according to task requirements.
2. Generate images using AI-based techniques	2.1 Effective prompts are developed based on the desired image outcome. 2.2 AI tools are used to generate images according to given instructions. 2.3 Generated images are adjusted for style, resolution, and format.
3. Process and enhance images using AI tools	3.1 <u>AI-assisted techniques</u> are applied for image enhancement and editing. 3.2 Image processing tasks such as background removal, upscaling, and style modification are performed. 3.3 Processed images are saved in appropriate file formats.
4. Apply quality and ethical standards in AI image generation	4.1 Generated images are checked for visual quality and accuracy. 4.2 Ethical guidelines and copyright considerations are followed during image generation. 4.3 Inappropriate or misleading AI-generated content is identified and avoided.
5. Organize and prepare AI-generated images for use	5.1 Generated and processed images are organized according to project requirements. 5.2 Images are properly named, stored, and documented. 5.3 Images are prepared for further use in design, media, or AI-related applications.

Range of Variables

Variable	Range May include but not limited to:
1. Different AI image generation tools and platforms	1.1 Text-to-image generation tools 1.2 Image-to-image generation tools 1.3 AI art generation platforms 1.4 AI photo enhancement tools 1.5 AI background generation tools 1.6 AI style transfer tools 1.7 AI-based design platforms 1.8 Cloud-based AI image tools 1.9 Open-source AI image generation tools 1.10 Web-based AI image generation platforms
2. AI-assisted techniques	2.1 AI-based image enhancement 2.2 AI background removal 2.3 AI object detection and replacement 2.4 AI image upscaling 2.5 AI noise reduction 2.6 AI color correction and colorization 2.7 AI style transfer 2.8 AI auto retouching 2.9 AI image restoration 2.10 AI generative fill techniques

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concept of AI-based image generation and processing tools used in digital design and media. 1.2 Identify different AI image generation platforms and their features. 1.3 Understand prompt development techniques for generating desired images. 1.4 Concept of AI image generation processes including style, resolution, and format adjustments. 1.5 Understand AI-assisted image editing and enhancement techniques. 1.6 Identify common image processing tasks such as background removal, upscaling, and style modification. 1.7 Procedures for saving and managing images in appropriate file formats. 1.8 Understand methods for organizing, naming, storing, and preparing images for design and media use.
2. Underpinning Skills	2.1 Selecting appropriate AI-based image generation and processing tools. 2.2 Reviewing and evaluating features and capabilities of AI

	image platforms. 2.3 Developing effective prompts to generate desired images using AI tools. 2.4 Using AI tools to generate images based on given instructions. 2.5 Applying AI-assisted techniques for image editing and enhancement. 2.6 Performing image processing tasks such as background removal, upscaling, and style modification. 2.7 Checking and ensuring visual quality and accuracy of generated images.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Learning materials 4.4 Specifications and work instructions 4.5 Relevant tools, equipment and facilities

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 identified AI-based image generation and processing 1.2 generated images using AI-based techniques 1.3 processed and enhanced images using AI tools 1.4 applied quality and ethical standards in AI image generation 1.5 organized and prepared AI-generated images for use
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or

	simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: IMPLEMENT INTERNAL CONTROL AND ENSURE DATA SECURITY	Nominal Duration: 10 hrs.	Unit Code: SICIP-ICT-AIPT-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to implement internal control and ensure data security. It specifically includes the tasks of applying image quality standards and guidelines, performing quality checking and error correction, following quality assurance procedures, maintaining documentation and version control and reporting quality issues and corrective actions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply image quality standards and guidelines	1.1 <u>Image quality standards and guidelines</u> are identified and reviewed. 1.2 Images are evaluated according to defined quality criteria. 1.3 Resolution, clarity, and visual consistency are checked against quality standards. 1.4 <u>Non-compliant images</u> are identified and corrected where required. 1.5 Final images are verified to ensure compliance with established quality guidelines.
2. Perform quality checking and error correction	2.1 <u>Image files</u> are reviewed to identify quality issues and errors. 2.2 Errors related to image clarity, color, and alignment are identified. 2.3 Necessary corrections are applied using <u>appropriate editing tools</u> . 2.4 Corrected images are rechecked to ensure quality standards are met. 2.5 Final outputs are verified for accuracy and consistency.
3. Follow quality assurance procedures	3.1 <u>Quality assurance procedures</u> are reviewed and followed during image processing tasks. 3.2 <u>Work outputs</u> are checked against established quality standards. 3.3 Identified issues or deviations are documented and reported. 3.4 Corrective actions are applied to address quality issues. 3.5 Final outputs are verified to ensure compliance with quality assurance requirements.
4. Maintain documentation and version control	4.1 Project documentation is prepared and maintained for

	image processing tasks. 4.2 File versions are recorded and managed according to version control procedures. 4.3 Changes and updates to image files are documented systematically. 4.4 Previous versions of files are preserved for reference and tracking. 4.5 Documentation and version records are reviewed to ensure accuracy and completeness.
5. Report quality issues and corrective actions	5.1 Quality issues in processed images are identified and documented. 5.2 Causes of identified quality problems are analyzed. 5.3 Corrective actions are recommended and implemented to resolve issues. 5.4 Quality issue reports are prepared and submitted to relevant stakeholders. 5.5 Implemented corrective actions are reviewed to ensure effectiveness.

Range of Variables

Variable	Range
	May include but not limited to:
1. Image quality standards and guidelines	1.1 Organizational image quality policies 1.2 Industry standards for digital imaging 1.3 Client or project-specific requirements 1.4 Platform-specific guidelines (web, print, social media)
2. Non-compliant images	2.1 Low resolution or pixelation 2.2 Blurry or out-of-focus images 2.3 Incorrect color representation 2.4 Overexposed or underexposed images 2.5 Inconsistent formatting or sizing
3. Image files	3.1 JPEG (JPG), PNG, TIFF, BMP formats 3.2 RAW image files from cameras 3.3 Scanned images 3.4 Web and print-ready images
4. Appropriate editing tools	4.1 Image editing software (e.g., Adobe Photoshop, GIMP, Lightroom) 4.2 Built-in adjustment tools (brightness, contrast, saturation) 4.3 Cropping, resizing, and rotation tools 4.4 Filters and correction presets
5. Quality assurance procedures	5.1 Organizational QA policies and standard operating procedures (SOPs) 5.2 Image processing workflows and checklists 5.3 Version control and file management procedures

	5.4 Documentation and reporting protocols
6. Work outputs	6.1 Edited digital images 6.2 Batch-processed image files 6.3 Web-ready and print-ready outputs 6.4 Image assets for multimedia or publications

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concept of image quality standards and guidelines in image processing work. 1.2 Identify quality criteria such as resolution, clarity, color, and consistency. 1.3 Understand procedures for quality checking and error correction in images. 1.4 Concept of common image errors and methods for correction using editing tools. 1.5 Understand quality assurance (QA) procedures and compliance requirements. 1.6 Concept of documentation and version control in digital image processing. 1.7 Understand maintaining file versions, change tracking, and record keeping. 1.8 Concept of reporting quality issues and evaluating corrective actions for continuous improvement.
2. Underpinning Skills	2.1 Applying image quality standards and evaluating images against defined criteria. 2.2 Checking and verifying resolution, clarity, and visual consistency of images. 2.3 Identifying and correcting image errors related to color, alignment, and quality. 2.4 Using appropriate editing tools to perform quality improvements. 2.5 Documenting and reporting quality issues and deviations from standards. 2.6 Applying corrective actions to resolve identified quality problems. 2.7 Maintaining documentation and version control for image files. 2.8 Reviewing and validating final outputs to ensure compliance and consistency.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn

	3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Specifications and work instructions 4.4 Relevant tools, equipment and facilities 4.5 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 applied image quality standards and guidelines 1.2 performed quality checking and error correction 1.3 followed quality assurance procedures 1.4 maintained documentation and version control 1.5 reported quality issues and corrective actions
2. Methods of Assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DEVELOP FREELANCING AND OUTSOURCING CAREER	Nominal Duration: 15 hrs.	Unit Code: SICIP-ICT-AIPT-07-O
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Unit Descriptor:

This unit of competency covers the knowledge, skills and attitudes required to develop freelancing and outsourcing career. It specifically includes the tasks of identifying freelancing and outsourcing market opportunities, developing professional profile and service portfolio, preparing proposals, gigs and acquire clients, managing client communication and long-term relationships and planning career growth and continuous professional development.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify freelancing and outsourcing market opportunities	1.1 <u>Freelancing and outsourcing markets</u> are researched to identify service demand. 1.2 Competition and pricing strategies in the market are analyzed. 1.3 Profitable niches and service areas are identified. 1.4 Required skills and competencies for outsourcing work are identified. 1.5 Client needs and market trends are reviewed. 1.6 Potential market opportunities are documented for service positioning.
2. Develop professional profile and service portfolio	2.1 A professional profile is created on relevant freelancing platforms. 2.2 Skills, qualifications, and work experience are presented in the profile. 2.3 A profile introduction video is prepared to present services and capabilities. 2.4 A service portfolio is developed to showcase completed work and skills. 2.5 Portfolio items and profile information are reviewed and updated regularly.
3. Prepare proposals, Gigs, and acquire clients	3.1 Service gigs are created and published on freelancing platforms. 3.2 Gig descriptions, pricing, and <u>service packages</u> are clearly defined. 3.3 Client profiles and ratings are reviewed before responding to job postings. 3.4 Job requirements are analyzed before preparing proposals. 3.5 Client proposals are prepared and submitted through

	appropriate freelancing platforms. 3.6 Client communications are maintained to secure projects and build trust.
4. Manage client communication and long-term relationships	4.1 Client communications are conducted through appropriate channels. 4.2 Client requirements and feedback are received and documented. 4.3 Project updates and service information are communicated regularly. 4.4 Client concerns or issues are addressed promptly. 4.5 Long-term client relationships are maintained through professional service and follow-up.
5. Plan career growth and continuous professional development	5.1 <u>Career goals and development needs</u> are identified and documented. 5.2 Required skills and competencies are reviewed against market demands. 5.3 Learning and training opportunities are identified for skill improvement. 5.4 <u>Professional development activities</u> are planned and scheduled.

Range of Variables

Variable	Range May include but not limited to:
1. Freelancing and outsourcing markets	1.1 Upwork 1.2 Fiverr 1.3 Freelancer 1.4 PeoplePerHour 1.5 Toptal 1.6 Guru 1.7 Truelancer 1.8 LinkedIn Marketplace / LinkedIn Services
2. Service packages	2.6 Basic service package 2.7 Standard service package 2.8 Premium service package 2.9 Custom service package 2.10 Hourly service package 2.11 Project-based service package 2.12 Monthly retainer package 2.13 Enterprise or advanced service package
3. Career goals and development needs	3.1 Short-term career goals 3.2 Long-term career goals 3.3 Technical skill development needs

	3.4 Industry certification goals 3.5 Training and learning priorities 3.6 Career progression or promotion goals 3.7 Personal development objectives
4. Professional development activities	4.1 Skills training and upskilling programs 4.2 Professional certification preparation 4.3 Networking and professional community participation 4.4 Knowledge sharing and peer learning 4.5 Industry research and self-study 4.6 Career planning and performance review activities

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concept of freelancing and outsourcing markets and global service demand. 1.2 Identify market opportunities, niches, and client needs in freelancing platforms. 1.3 Understand competition, pricing strategies, and market trends. 1.4 Concept of professional profiles and service portfolios for online platforms. 1.5 Identify client requirements and project expectations before submitting proposals. 1.6 Concept of client communication and relationship management in freelancing. 1.7 Understand strategies for building long-term client relationships and trust. 1.8 Concept of career planning and continuous professional development.
2. Underpinning Skills	2.1 Researching and analyzing freelancing and outsourcing markets to identify opportunities. 2.2 Evaluating competition, pricing strategies, and profitable service niches. 2.3 Developing and presenting professional profiles and service portfolios. 2.4 Analyzing client requirements and preparing effective proposals. 2.5 Communicating and negotiating with clients to secure projects. 2.6 Maintaining and updating portfolio, profile, and project records. 2.7 Planning and pursuing continuous professional development and career growth.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety

	3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Specifications and work instructions 4.4 Relevant tools, equipment and facilities 4.5 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 identified freelancing and outsourcing market opportunities 1.2 developed professional profile and service portfolio 1.3 prepared proposals, gigs, and acquire clients 1.4 managed client communication and long-term relationships 1.5 planned career growth and continuous professional development
2. Methods of Assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Advanced Image Processing & Technique
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom cum workshop – 500 sft

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptop	01
2.	Printer	01
3.	Scanner	01
4.	Computers/Laptop: (Minimum Intel Core i5 processor (8 th Gen), 8GB RAM (16GB preferred), 256GB SSD/HDD storage, High-resolution monitors (19/20-inch minimum, with color accuracy).	25
5.	Projector: High-resolution projector (HD or 4K) for instructor presentations, large screen for the room to ensure visibility for all trainees.	01
6.	High-Speed Internet Router	01
7.	LAN Networking System (Cables and Ports)	01
8.	UPS (Uninterruptible Power Supply)	25
9.	External Hard Drive / Backup Storage	02
10.	Projection Screen	01
11.	Whiteboard / Smart Board	01
12.	Computer Tables	25
13.	Computer Chairs	25
14.	Trainer Table and Chair	02

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency

standards (CS) provided by SICIP.

For the operation of training course on Advanced Image Processing & Technique, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.